Repurposing the Tangram task to study (in)definiteness in the absence of definites

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Research on the production and comprehension of referring expressions, especially when considered against a visual context, has largely focused on definite descriptions (i.e., those involving '*the*', which is also the most frequently used word in English). Yet, there are in fact more languages in the world that do not have an overt definite determiner (62.5%) [1]. For instance, in Mandarin Chinese, indefinite meanings are expressed with *one*-classifier constructions, while definite meanings involve bare nominals and demonstrative descriptions [2,3] (Table 1).

How can we study referring expressions in a way that is, on the one hand, naturalistic, and, at the same time, provides some control over the meanings expressed? We reasoned that this can be done with the classic Tangram task, where a Director instructs a Matcher to order and reorder a set of abstract images (Fig. 1) [4,5]. There is extensive evidence that the referring expressions produced in this task (i) get shorter over time; (ii) create a conceptualization for each tangram; (iii) move from indefinites to definites [6,7,8]. Instead of using these measures to study collaboration and common ground, we focus on the reliability of the indefinite-to-definite progression to study **referential forms** in Mandarin. We directly manipulate *uniqueness* (Unique: 6 mammals, Non-unique: 6 birds, Fig. 2); *familiarity* is inherently part of the task, with re-mention over 6 rounds: these are the two hallmarks of definiteness.

RESULTS (n=14) (Fig. 3)**.** We replicate the two other known effects [4,5]: (i) **increased efficiency:** over time, in mean number of words (83, 41, 33, 33, 18, 17); (ii) convergence on **conceptualizations**: We quantified this by extracting triplets of the most frequent content words per tangram per dyad (nouns, adjectives, verbs): each dyad used a different triplet for each of the 12 tangrams. These replications allow us to assume that the indefinite-to-definite meaning progression holds in this data. We consider three aspects of referential forms.

Grammaticalization of labels. One aspect not considered in previous studies is the grammatical position of the "target" head noun. We distinguish head N (*"the skinny dog"*) from Ns inside a property ("*looks like a dog"*): head Ns increases over rounds (β =0.66, p < .001). There was also an interaction between Round and Uniqueness: speakers are quicker to converge on a head noun in the uniqueness condition (e.g., *"dog"*) compared to non-uniqueness (*"bird"*) (β =-0.53, p < .001).

Modification. We also distinguish modifiers that occur in their canonical position in Mandarin, which is **before** the noun, from post-nominal modification which is an elaboration. Overall, canonical pre-nominal modification is *less* likely for unique compared to non-unique referents (β = -1.24, p < .005). There was also a significant interaction between Rounds and Uniqueness (β = 0.23, p < .005): prenominal modification increases at a slower rate for unique referents, considering that these could be distinguished with the noun alone (in contrast to birds).

Determiners: one, bare, demonstrative. For the overt indefinite one-classifier (yellow), we find the expected decrease of over time (β = -1.48, p < .001), but, in addition, indefinites are overall less likely for referents in the unique condition (β = 1.53, p = .008). For bare nominals (blue), they are overall produced more for referents in unique (73% vs. 51%, β = -1.24, p = .009). They also increase over Rounds (β = 0.84, p < .001): for referents in Unique, bare nominals increase by 13.4% per round, and 10.2% for referent in Non-Unique referents. This suggests that bare nominals in Mandarin are the preferred referential strategy for what in English would surface as definites. For demonstrative descriptions (red), they do not differ across uniqueness conditions (p = .66), although Uniqueness does interact with Round: demonstratives are more likely for referents in Non-unique over time (β = 0.44, p = .006). This suggests that Mandarin demonstrative descriptions do not generally function as definite descriptions.

CONCLUSION. We leverage the well-established indefinite-to-definite meaning progression in the classic tangram task to examine referential strategies in Mandarin, a language without a definite determiner. Replicating other known effects in this task, efficiency and conceptualization, allows us to study definite meanings. We find that in Mandarin, bare nominals are the preferred

strategy, and not demonstrative descriptions. The novel patterns observed here contribute to our understanding of reference cross-linguistically, and can guide research in other languages.

Table 1: **The Mandarin Noun Phrase**	
yi zhi huanle DE gou one CL happy DE dog ‘ (a/one) happy dog’	One-Classifier (Indefinite)
na zhi huanle DE gou DEM CL happy DE dog ‘that happy dog’	Demonstrative
huanle DE gou happy DE dog ‘happy dog(s)’	Bare nominal

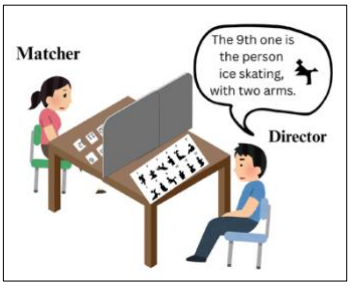
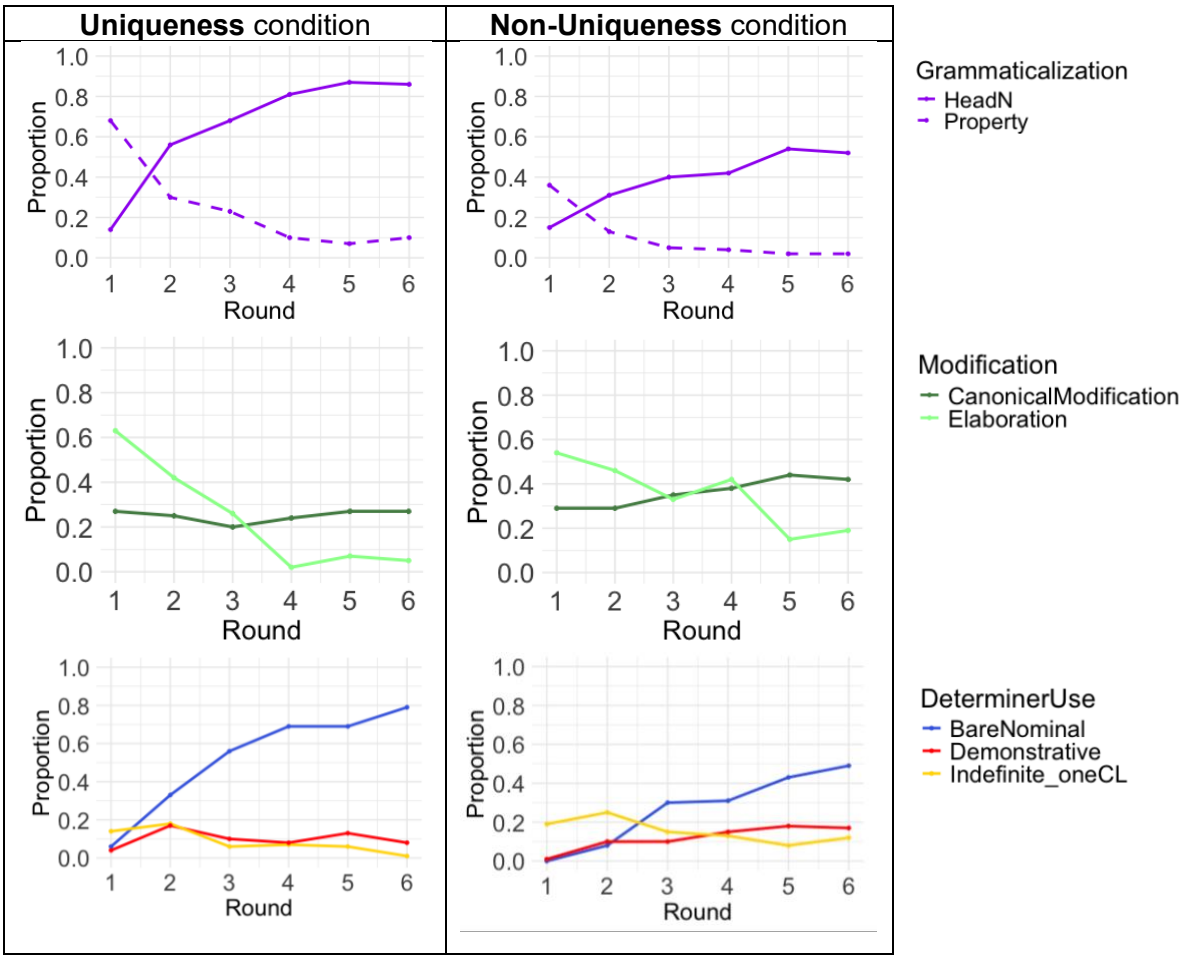


Fig. 1. The Tangram task. [4] Over time, **indefinites** (“11 looks like **an** ice skater”) move to **definites** “4 is **the** ice skater”)

Fig. 2. Tangrams adapted from Duff et al. (2012) [7], split into two conditions: *Uniqueness* (six highly discriminable mammals (e.g., camel, dog)); and *Non-Uniqueness* (six lowly discriminable birds).

Fig. 3. Tangram results by condition, organized by grammaticalization, modification and determiner-use.



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